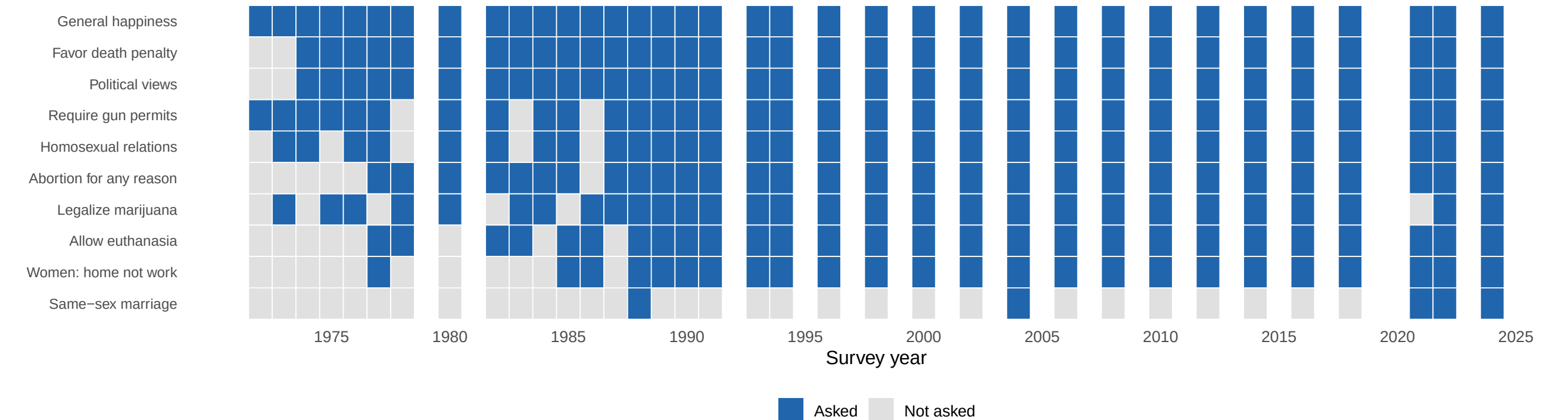


Missing Data Mechanisms in the General Social Survey

Panels A–D: Standard MCAR/MNAR diagnostics inspired by Kim & Lee (2024). Panels E–F: NOVEL design-aware analysis using the survey package -- showing that ignoring the GSS complex design (stratification, clustering, weights) compounds the bias from MNAR missingness.

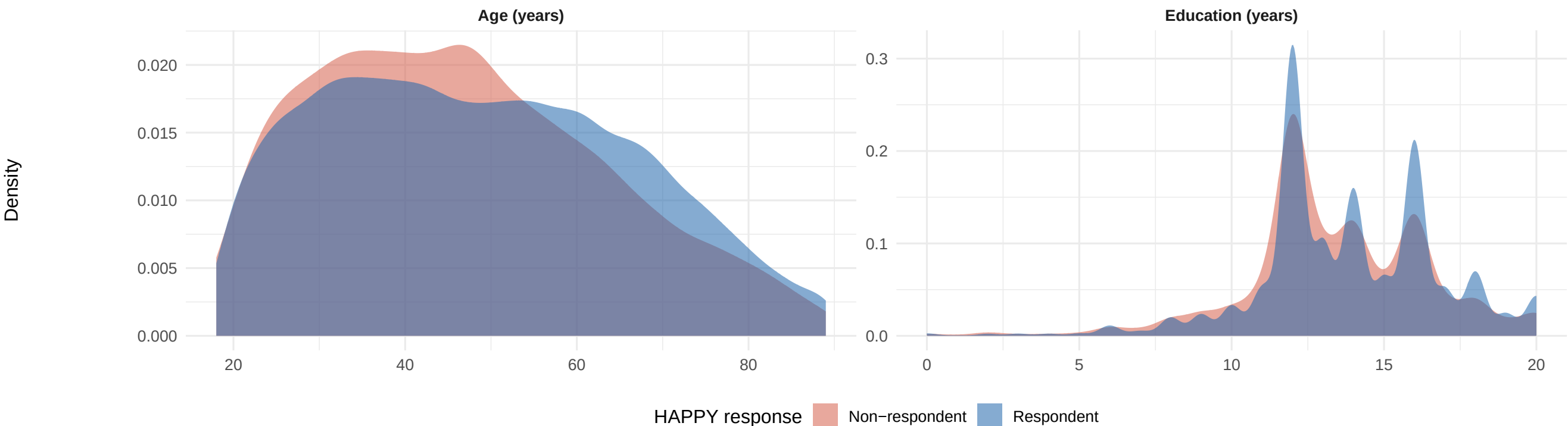
A. Structural Missingness by Survey Design (MNAR)

GSS opinion questions are only asked in certain years -- missingness is NOT random



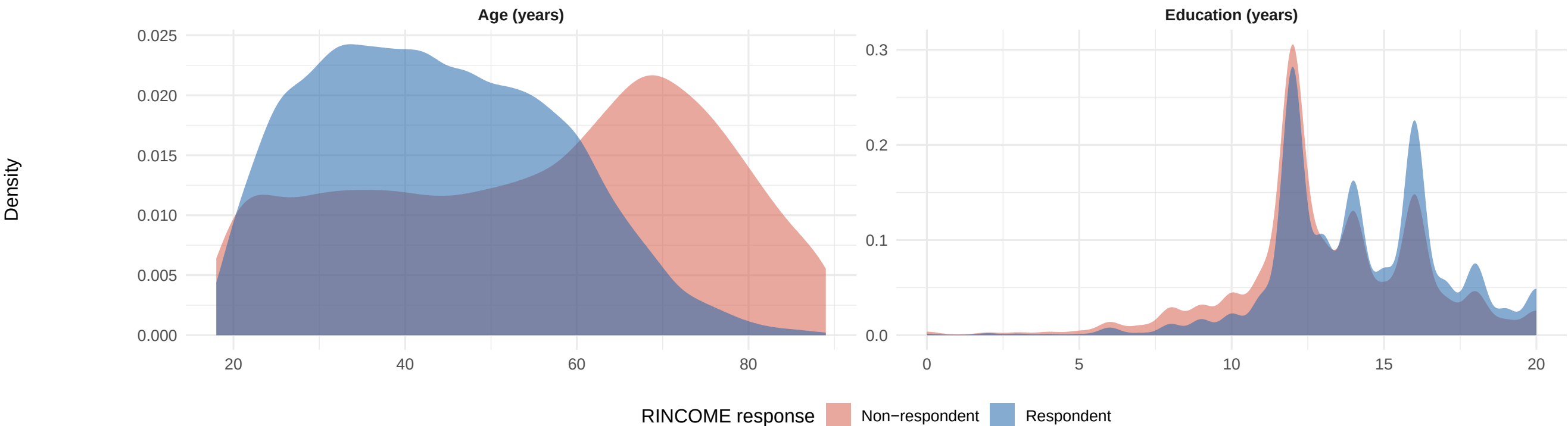
B. MCAR Diagnostic: 'General Happiness' Item Non-Response

Distributions nearly identical -- missingness is unrelated to demographics (MCAR)



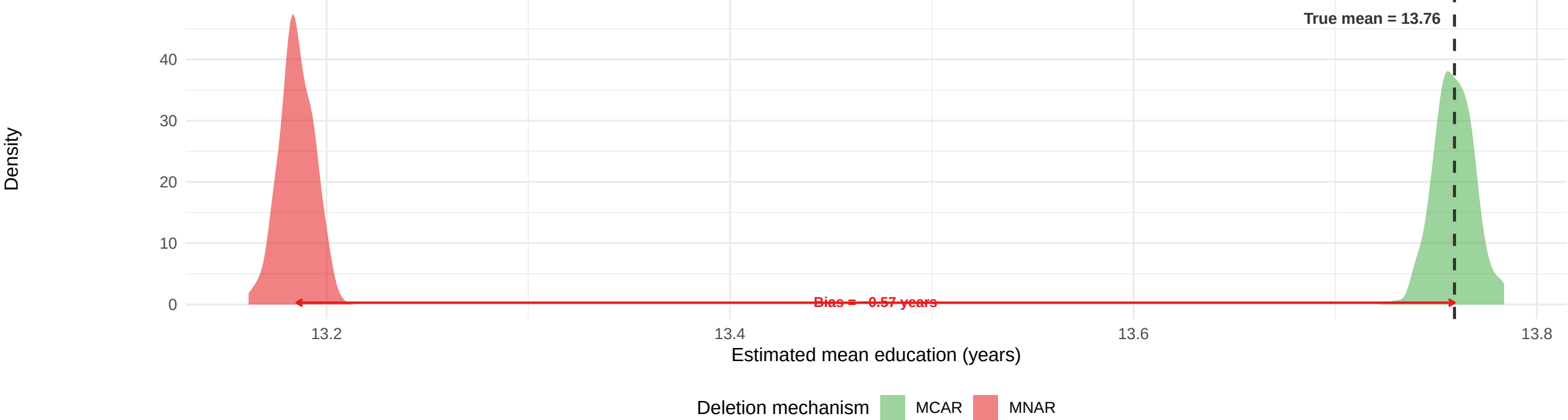
C. MNAR Diagnostic: Respondent Income Non-Response

Distributions differ systematically -- missingness depends on unobserved income (MNAR)



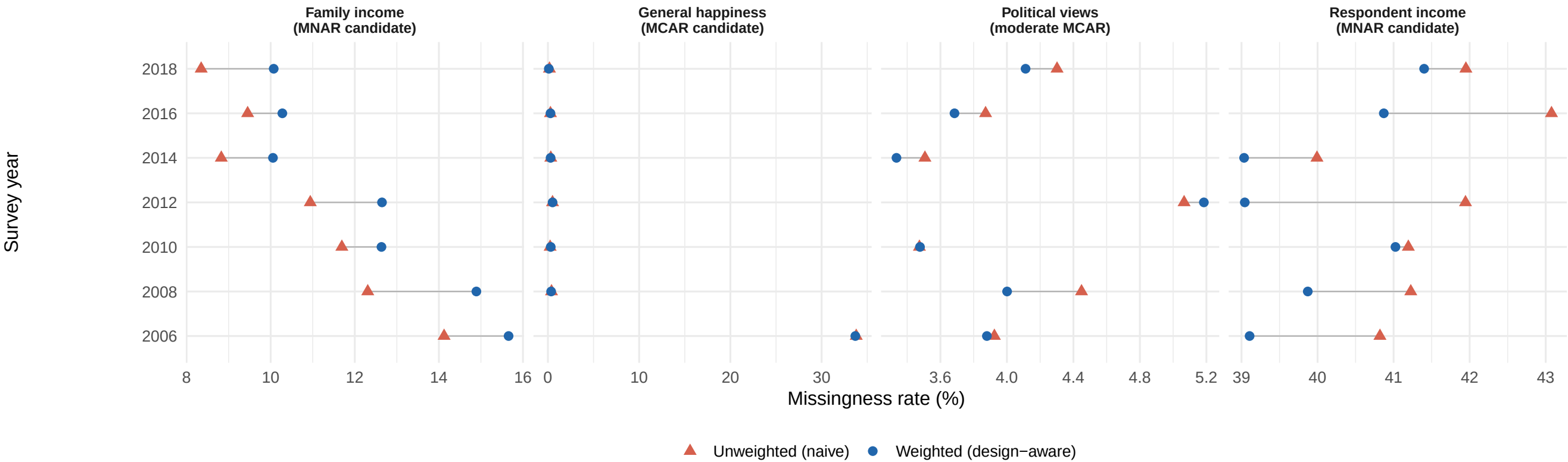
D. Estimation Bias: MCAR Preserves Truth, MNAR Distorts It

200 simulations of 30% deletion on GSS education data (n = 37,415)



E. Novel Diagnostic: Weighted vs Unweighted Missingness Rates

If MCAR, weighted and unweighted rates coincide (left panels). Under MNAR, they diverge (right panels) because survey weights correlate with the missing mechanism. Per-wave analysis 2006–2018; survey.lonely.psu = 'adjust'.



F. Novel Finding: Design-Ignorance Compounds MNAR Bias

Mean education (years) in GSS 2018 estimated three ways. Under MCAR (left), all methods agree. Under MNAR (right), ignoring both the design AND the missing mechanism produces the largest error. Error bars: 95% CI.

